

The background is a deep blue gradient with a subtle pattern of white dots. Overlaid on the left side are several concentric circles and a large circular scale with degree markings from 140 to 260. Some circles have arrows indicating a clockwise direction. The text is centered on the right side of the image.

ARTIFICIAL INTELLIGENCE

UNIT 1

INTELLIGENCE

- the ability to solve complex problems or make decisions with outcomes benefiting the actor
- Intelligence is a general mental ability that integrates cognitive functions like perception, attention, memory, language, and planning
- **the ability to learn from experience, acquire and retain knowledge, and use reason to solve problems.**

ARTIFICIAL INTELLIGENCE

- computer systems capable of performing complex tasks that historically only a human could do, such as reasoning, making decisions, or solving problems.
- Artificial intelligence is the simulation of human intelligence processes by machines, especially computer systems. Specific applications of AI include expert systems, natural language processing, speech recognition and machine vision.

EXAMPLES

- ChatGPT: Uses large language models (LLMs) to generate text in response to questions or comments posed to it.
- Google Translate: Uses deep learning algorithms to translate text from one language to another.
- Netflix: Uses machine learning algorithms to create personalized recommendation engines for users based on their previous viewing history.
- Tesla: Uses computer vision to power self-driving features on their cars.

4 MAIN TYPES OF ARTIFICIAL INTELLIGENCE

1. Reactive machines

- Reactive machines are AI systems that have no memory and are task specific, meaning that an input always delivers the same output. Machine learning models tend to be reactive machines because they take customer data, such as purchase or search history, and use it to deliver recommendations to the same customers.
- This type of AI is reactive. It performs “super” AI, because the average human would not be able to process huge amounts of data such as a customer’s entire Netflix history and feedback customized recommendations

2. Limited memory machines

- The next type of AI in its evolution is limited memory. This algorithm imitates the way our brains' neurons work together, meaning that it gets smarter as it receives more data to train on. Deep learning algorithms improve natural language processing (NLP), image recognition, and other types of reinforcement learning.
- **Self-driving cars:** A good example of limited memory AI is the way self-driving cars observe other cars on the road for their speed, direction, and proximity. This information is programmed as the car's representation of the world, such as knowing traffic lights, signs, curves, and bumps in the road.

3. Theory of mind

- The first two types of AI, reactive machines and limited memory, are types that currently exist. Theory of mind and self-aware AI are theoretical types that could be built in the future. As such, there aren't any real world examples yet.
- If it is developed, theory of mind AI could have the potential to understand the world and how other entities have thoughts and emotions. In turn, this affects how they behave in relation to those around them.

4. Self-awareness

- The grand finale for the evolution of AI would be to design systems that have a sense of self, a conscious understanding of their existence. This type of AI does not exist yet.
- This goes a step beyond theory of mind AI and understanding emotions to being aware of themselves, their state of being, and being able to sense or predict others' feelings. For example, "I'm hungry" becomes "I know I am hungry" or "I want to eat paneer tikka because it's my favorite food."

CHALLENGES

Artificial intelligence (AI) faces many challenges, including:

- Data privacy

AI often involves processing large amounts of sensitive and confidential information, which can raise concerns about data security and privacy. Hackers and cybercriminals can exploit vulnerabilities in AI systems to gain access to sensitive information.

- Bias

AI systems are only as good as the data they are trained on, and if that data is biased, the AI system will be biased as well. Bias can lead to unfair treatment of certain groups of people and can perpetuate existing inequalities.

- Ethical challenges

As AI systems become increasingly autonomous, questions arise about accountability, bias, and privacy. Ensuring that AI operates in a fair and responsible manner remains a pressing concern.

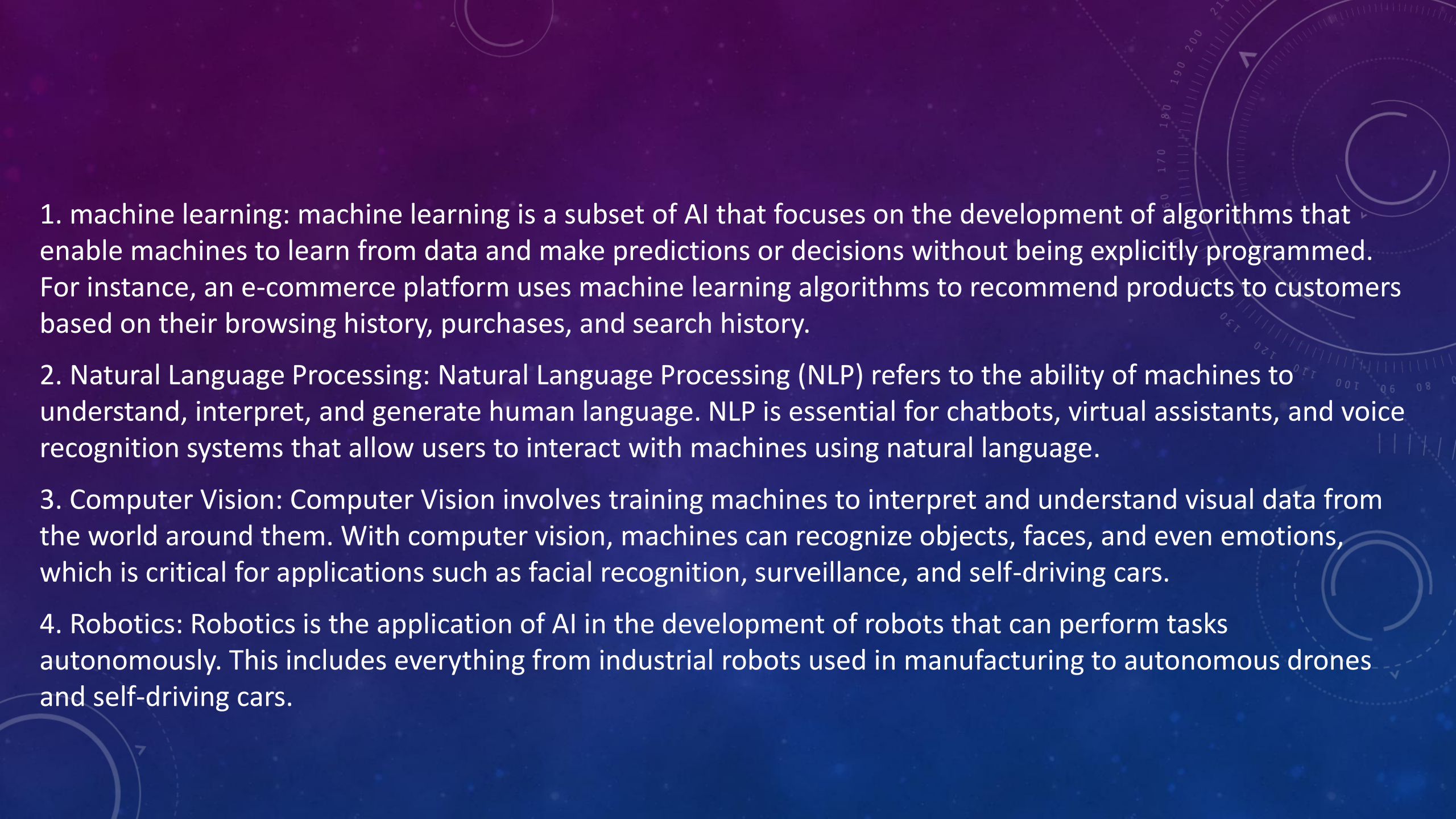
- Data quality

AI systems rely heavily on the data they are fed, and if the data is inaccurate, incomplete, or biased, the results of the AI will also be flawed.

THE FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

The Foundation of AI

These building blocks include **machine learning, natural language processing, computer vision, and robotics**, among others. Together, these components form the backbone of AI, allowing machines to learn, adapt, and improve over time

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1. machine learning: machine learning is a subset of AI that focuses on the development of algorithms that enable machines to learn from data and make predictions or decisions without being explicitly programmed. For instance, an e-commerce platform uses machine learning algorithms to recommend products to customers based on their browsing history, purchases, and search history.
 2. Natural Language Processing: Natural Language Processing (NLP) refers to the ability of machines to understand, interpret, and generate human language. NLP is essential for chatbots, virtual assistants, and voice recognition systems that allow users to interact with machines using natural language.
 3. Computer Vision: Computer Vision involves training machines to interpret and understand visual data from the world around them. With computer vision, machines can recognize objects, faces, and even emotions, which is critical for applications such as facial recognition, surveillance, and self-driving cars.
 4. Robotics: Robotics is the application of AI in the development of robots that can perform tasks autonomously. This includes everything from industrial robots used in manufacturing to autonomous drones and self-driving cars.

TECHNIQUES

Artificial intelligence (AI) techniques can be categorized into three types: **Supervised learning, Unsupervised learning, Reinforcement learning.**

Each technique has its own strengths and weaknesses, and can be applied to a wide range of problems and applications

- **Supervised Learning:** The algorithm is trained on a labeled dataset, where the input-output pairs are provided. The algorithm learns the relationship between input and output and applies this knowledge to unseen data. Examples: Linear Regression, Support Vector Machines (SVM), and Neural Networks.
- **Unsupervised Learning:** The algorithm is provided with an unlabeled dataset, and it identifies patterns or structures in the data without guidance. Examples: Clustering (e.g., K-means), Dimensionality Reduction (e.g., Principal Component Analysis), and Association Rule Learning.
- **Reinforcement Learning:** The algorithm learns from its actions and interactions with an environment to maximize a reward signal. It's particularly useful in decision-making and control tasks. Examples: Q-learning, Deep Q-Network (DQN), and Policy Gradient methods.

MODELS

Artificial intelligence (AI) models are essential for AI to **learn how to perform tasks** like facial recognition, email spam detection, and product recommendations. AI models **require a dataset to learn** how to perform these tasks

Artificial intelligence models are the **tools and algorithms used to train computers** to process and analyze data – just as humans do

- An AI model is a program that has been trained on a set of data to recognize certain patterns or make certain decisions without further human intervention. Artificial intelligence models apply different algorithms to relevant data inputs to achieve the tasks, or output, they've been programmed for.

DEFINING PROBLEM AS A STATE SPACE SEARCH

- A state space is a mathematical representation of a problem that defines **all possible states** that the problem can be in. Furthermore, in search algorithms, we use a state space to represent the **current state of the problem, the initial state, and the goal state.**

- The state space search in artificial intelligence is a search algorithm that is used in computer science. Artificial intelligence is a process by which the search algorithm is desired to find the desired goal from all the probabilities. The states and the instances are considered to find the goal state within the state space process.
- The state space search algorithm is described as the set of ordered pairs with two integers.

- State: It can be the Initial State, the Goal State, or any state generated by applying rules.
- Space: In AI, Space consists of the complete set of conceivable states for a given problem.
- Search: This technique progresses from the initial state to the desired state by applying effective rules within the space of all possible states.
- Search Tree: It's a visual representation of the problem, starting with the initial state as the root node.
- Transition Model: Describes the impact of each action. Path Cost assigns a value to each sequence connecting start and end nodes, with the optimal option having the lowest cost.

STATE SPACE SEARCH IS A FUNDAMENTAL TECHNIQUE IN ARTIFICIAL INTELLIGENCE WITH VARIOUS APPLICATIONS ACROSS DIFFERENT DOMAINS

- **Puzzle Solving:** Solving puzzles like the 8-puzzle, Rubik's Cube, and Sudoku using state space search algorithms
- **Pathfinding in Games:** Finding the shortest path for characters or agents in video games is a common use case for algorithms like A*
- **Robotics:** Planning the movement of robots in a physical environment to perform tasks or reach specific locations
- **Automated Planning:** In areas like logistics, transportation, and manufacturing, state space search helps in planning and scheduling tasks
- **Natural Language Processing:** In tasks like machine translation, state space search can be used to generate optimal translations
- **Chess and Games:** Determining optimal moves in games with well-defined rules and states, like chess, checkers, and Go
- **Optimization Problems:** Solving optimization problems in areas like resource allocation, scheduling, and financial modeling

PRODUCTION SYSTEM

- A production system in AI is a computer-based system designed to automate decision-making and problem-solving tasks. It comprises a global database, production rules, and a control system to process data and derive conclusions
- AI Production Systems are the backbone of decision-making. These systems automate complex tasks through production rules, efficiently processing data and generating insights. They facilitate knowledge-intensive processes comprising a global database, production rules, and a control system. Their key features include simplicity, modularity, adaptability, and modifiability

The components of an AI Production System encompass three essential elements:

- **Global Database:** The global database serves as the system's memory, storing facts, data, and knowledge relevant to its operation. It is a repository that production rules can access to make informed decisions and draw conclusions.
- **Production Rules:** Production rules form the core logic of the system. They are a set of guidelines that the system follows while making decisions. These regulations outline the system's reaction to various inputs and circumstances.
- **Control System:** The control system manages the execution of production rules. It determines the sequence in which rules are applied, ensuring efficient processing and optimizing the system's performance.

Production System Rules

- Production system rules are the fundamental building blocks of AI systems. These rules define the logic and actions that guide the system's decision-making process.
- In an AI production system, rules encode knowledge and specify how the system should respond to different inputs and conditions. Production rules consist of conditions (if part) and actions (then part), which are applied based on the system's current state and available data.

REAL-WORLD EXAMPLES OF AI PRODUCTION SYSTEMS IN USE

- **Customer Support Chatbots:** AI-powered chatbots in customer support systems use production rules to handle customer inquiries, provide answers, and escalate complex issues to human agents.
- **Fraud Detection Systems:** In financial institutions, AI production systems detect fraudulent activities by analyzing transaction data and applying predefined fraud detection rules.
- **Medical Diagnosis:** AI production systems are used in healthcare for medical diagnosis. They analyze patient symptoms, medical history, and test results to suggest possible diagnoses and treatment options.
- **Traffic Management:** Smart traffic management systems use AI production systems to optimize traffic flow by adjusting signal timings based on real-time traffic conditions and predefined rules.

FEATURES OF A PRODUCTION SYSTEM IN AI

- **Simplicity:** Production Systems offer a straightforward way to encode and execute rules, making them accessible for developers and domain experts.
- **Modularity:** These systems are composed of modular components, allowing for the addition, removal, or modification of rules without disrupting the entire system. This modularity enhances flexibility and ease of maintenance.
- **Modifiability:** AI Production Systems are highly adaptable. Rules can be updated or replaced without extensive reengineering, ensuring the system remains up-to-date and aligned with evolving requirements.
- **Knowledge-intensive:** They excel in handling knowledge-rich tasks, relying on a comprehensive global database.
- **Adaptability:** AI Production Systems can dynamically adapt to new data and scenarios. This adaptability allows them to continuously improve.

CLASSIFICATION OF PRODUCTION SYSTEMS IN AI

AI production systems can be classified into four common classifications:

- **Monotonic Production System:** In a monotonic production system, the laws and truths remain constant while being carried out. A rule remains constant throughout the procedure once a fact is deduced. This stability ensures predictability but may limit adaptability in dynamic environments.
- **Partially Commutative Production System:** In this type of system, rules can be applied flexibly, allowing for some degree of adaptability while maintaining certain constraints. Partial commutativity strikes a balance between stability and flexibility.
- **Non-monotonic Production System:** Non-monotonic production systems are more dynamic and adaptive. Rules can be added, modified, or retracted during execution. They are excellent for situations where the knowledge base needs to change in response to shifting circumstances because of their flexibility.
- **Commutative System:** Commutative systems have rules that can be applied in any sequence without changing the result. In circumstances where the sequence of rule application is not essential, this high degree of flexibility may be beneficial.

INTELLIGENT AGENTS

- An AI system is composed of an agent and its environment. The agents act in their environment. The environment may contain other agents.

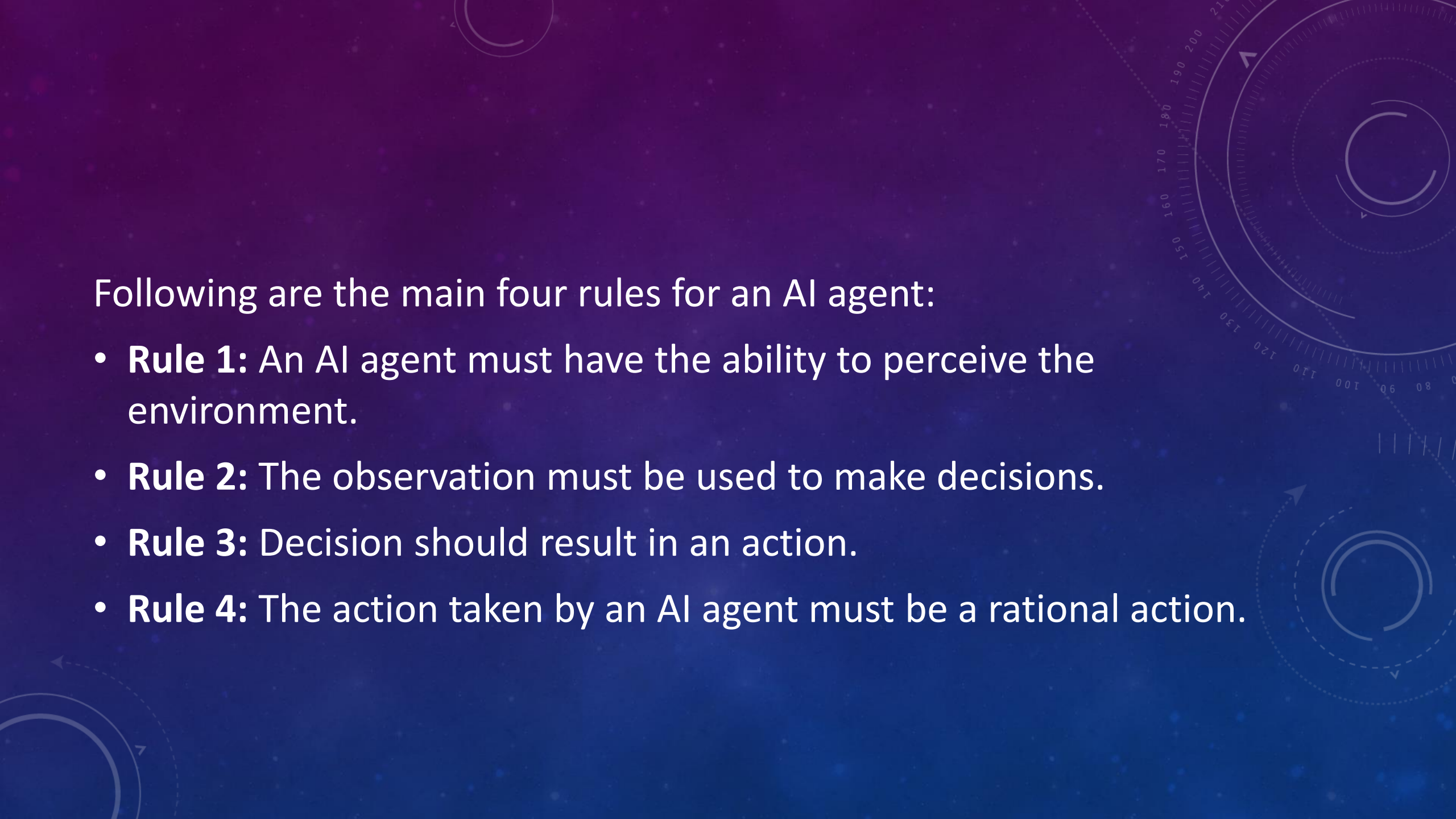
An agent is anything that can be viewed as:

- Perceiving its environment through **sensors** and
- Acting upon that environment through **actuators**(actuators are the physical or digital mechanisms that an AI agent uses to carry out actions in its environment. They bridge the gap between the agent's decision-making process and its impact on the surroundings.)

- an agent is a computer program or system that is designed to perceive its environment, make decisions and take actions to achieve a specific goal or set of goals. The agent operates autonomously, meaning it is not directly controlled by a human operator.
- An intelligent agent (IA) is a program that acts independently to achieve goals by using sensors and actuators to perceive its environment. IAs can be robots, machines, humans, or animals. They can learn from their environment to achieve their goals.

Some examples of intelligent agents include:

- Driverless cars
- Siri virtual assistant
- Software agents



Following are the main four rules for an AI agent:

- **Rule 1:** An AI agent must have the ability to perceive the environment.
- **Rule 2:** The observation must be used to make decisions.
- **Rule 3:** Decision should result in an action.
- **Rule 4:** The action taken by an AI agent must be a rational action.

STRUCTURE OF AN AI AGENT

The task of AI is to design an agent program which implements the agent function. The structure of an intelligent agent is a combination of architecture and agent program. It can be viewed as:

- Agent = Architecture + Agent program
- Following are the main three terms involved in the structure of an AI agent:
- **Architecture:** Architecture is machinery that an AI agent executes on.
- **Agent Function:** Agent function is used to map a percept to an action.
- **Agent program:** Agent program is an implementation of agent function. An agent program executes on the physical architecture to produce function f.

PEAS Representation

- PEAS is a type of model on which an AI agent works upon. When we define an AI agent or rational agent, then we can group its properties under PEAS representation model. It is made up of four words:
- **P:** Performance measure
- **E:** Environment
- **A:** Actuators
- **S:** Sensors

self-driving car then PEAS representation will be:

- **Performance:** Safety, time, legal drive, comfort
- **Environment:** Roads, other vehicles, road signs, pedestrian
- **Actuators:** Steering, accelerator, brake, signal, horn
- **Sensors:** Camera, GPS, speedometer, odometer, accelerometer, sonar.

- Agents can be classified into different types based on their characteristics, such as whether they are reactive or proactive, whether they have a fixed or dynamic environment, and whether they are single or multi-agent systems.
- Reactive agents are those that respond to immediate stimuli from their environment and take actions based on those stimuli. Proactive agents, on the other hand, take initiative and plan ahead to achieve their goals. The environment in which an agent operates can also be fixed or dynamic. Fixed environments have a static set of rules that do not change, while dynamic environments are constantly changing and require agents to adapt to new situations.
- Multi-agent systems involve multiple agents working together to achieve a common goal. These agents may have to coordinate their actions and communicate with each other to achieve their objectives. Agents are used in a variety of applications, including robotics, gaming, and intelligent systems. They can be implemented using different programming languages and techniques, including machine learning and natural language processing.

- A rational agent could be anything that makes decisions, such as a person, firm, machine, or software. It carries out an action with the best outcome after considering past and current percepts

Architecture and Agent programs

Architecture is the machinery that the agent executes on. It is a device with sensors and actuators, for example, a robotic car, a camera, and a PC. **An agent program** is an implementation of an agent function.

An **agent function** is a map from the percept sequence(history of all that an agent has perceived to date) to an action.

CHARACTERISTICS

Agents in Artificial Intelligence contain the following Characteristics:

- Environment
- Autonomous
- Flexibility
- Reactive
- Proactiveness
- Using Response Rules

- Environment

The agent is situated in a given environment.

- Autonomous

The agent can operate without direct human intervention or other software methods. It controls its activities and internal environment. The agent independently chooses the steps it will take in its current condition to achieve the best improvements. The agent achieves autonomy if its performance is measured by its experiences in the context of learning and adapting.

Flexibility

- Reactive: Agents must recognize their surroundings and react to the changes within them.
- Proactive: Agents shouldn't only act in response to their surroundings but also be able to take the initiative when appropriate and effect an opportunistic, goal-directed performance.
- Social: Agents should work with humans or other non-human agents.

Reactive

- Reactive systems maintain ongoing interactions with their environment, responding to its changes.
- The program's environment may be guaranteed, not concerned about its success or failure.
- Most environments are dynamic, meaning that things are constantly in a state of change, and information is incomplete.
- Programs must make provisions for the possibility of failure.

Pro-Activeness

- Taking the initiative to create goals and try to meet them.

Using Response Rules

- The goal for the agent is directed behavior, having it do things for the user.
- Mobility: The agent must have the ability to actuate around a system.
- Veracity: If an agent's information is false, it will not communicate.
- Benevolence: Agents don't have contradictory or conflicting goals. Therefore, every Agent will always try to do what it is asked.
- Rationality: The agent will perform to accomplish its goals and not work in a way that opposes or blocks them.
- Learning: An agent must be able to learn.

SEARCH METHODS AND ISSUES IN THE DESIGN OF SEARCH PROBLEMS

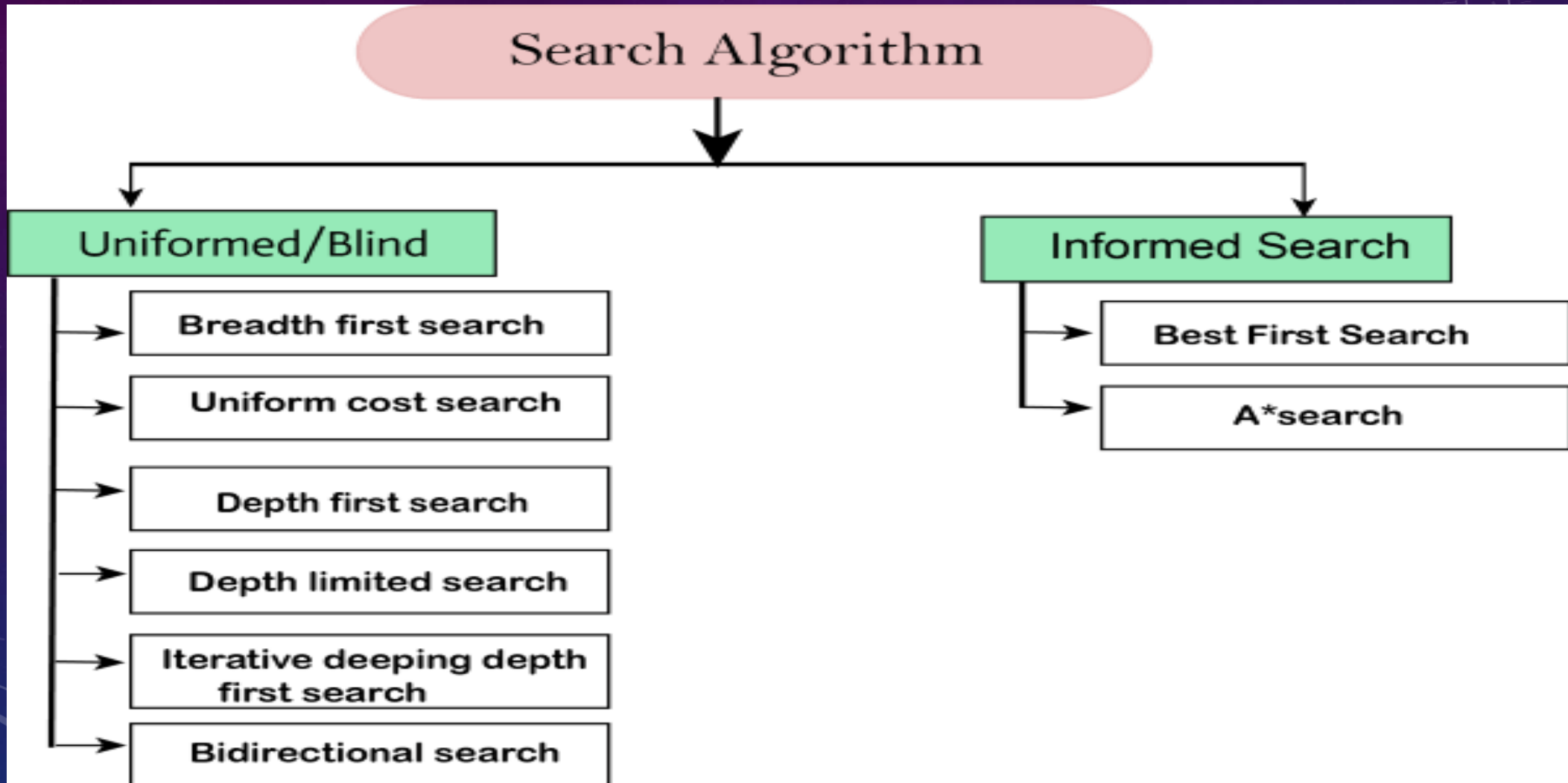
- **Search:** Searching is a step by step procedure to solve a search-problem in a given search space. A search problem can have three main factors:
 - **Search Space:** Search space represents a set of possible solutions, which a system may have.
 - **Start State:** It is a state from where agent begins **the search**.
 - **Goal test:** It is a function which observe the current state and returns whether the goal state is achieved or not.

- **Search tree:** A tree representation of search problem is called Search tree. The root of the search tree is the root node which is corresponding to the initial state.
- **Actions:** It gives the description of all the available actions to the agent.
- **Transition model:** A description of what each action do, can be represented as a transition model.
- **Path Cost:** It is a function which assigns a numeric cost to each path.
- **Solution:** It is an action sequence which leads from the start node to the goal node.
- **Optimal Solution:** If a solution has the lowest cost among all solutions.

PROPERTIES OF SEARCH ALGORITHMS

- **Completeness:** A search algorithm is said to be complete if it guarantees to return a solution if at least any solution exists for any random input.
- **Optimality:** If a solution found for an algorithm is guaranteed to be the best solution (lowest path cost) among all other solutions, then such a solution for is said to be an optimal solution.
- **Time Complexity:** Time complexity is a measure of time for an algorithm to complete its task.
- **Space Complexity:** It is the maximum storage space required at any point during the search, as the complexity of the problem.

TYPES OF SEARCH ALGORITHMS



- Uninformed/Blind Search:

The uninformed search does not contain any domain knowledge such as closeness, the location of the goal. It operates in a brute-force way as it only includes information about how to traverse the tree and how to identify leaf and goal nodes. Uninformed search applies a way in which search tree is searched without any information about the search space like initial state operators and test for the goal, so it is also called blind search. It examines each node of the tree until it achieves the goal node.

Informed Search

- Informed search algorithms use domain knowledge. In an informed search, problem information is available which can guide the search. Informed search strategies can find a solution more efficiently than an uninformed search strategy. Informed search is also called a Heuristic search.
- A heuristic is a way which might not always be guaranteed for best solutions but guaranteed to find a good solution in reasonable time.

BREADTH-FIRST SEARCH

- Breadth-first search is the most common search strategy for traversing a tree or graph. This algorithm searches breadthwise in a tree or graph, so it is called breadth-first search.
- BFS algorithm starts searching from the root node of the tree and expands all successor node at the current level before moving to nodes of next level.
- Uses FIFO queue

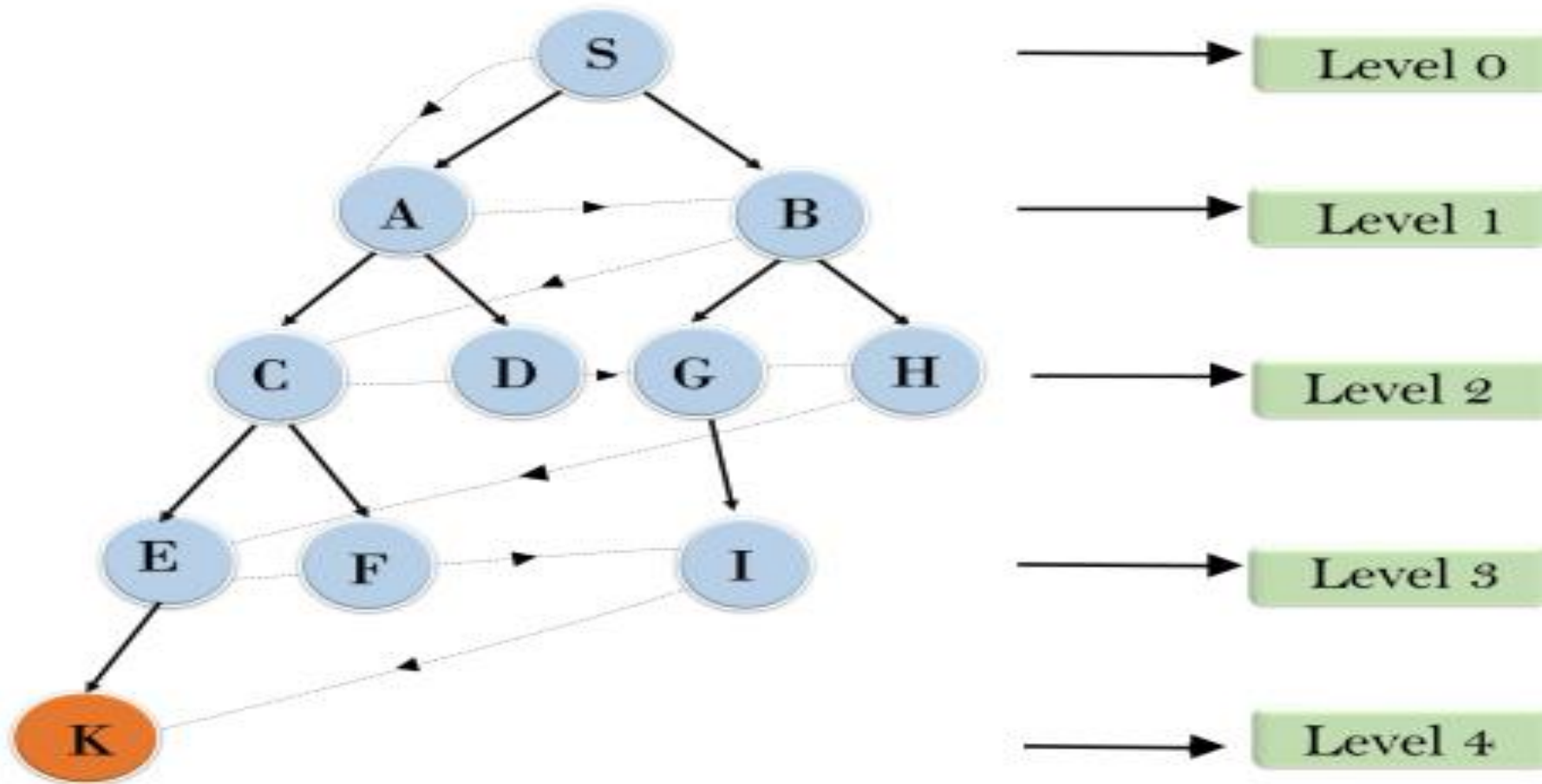
Advantages:

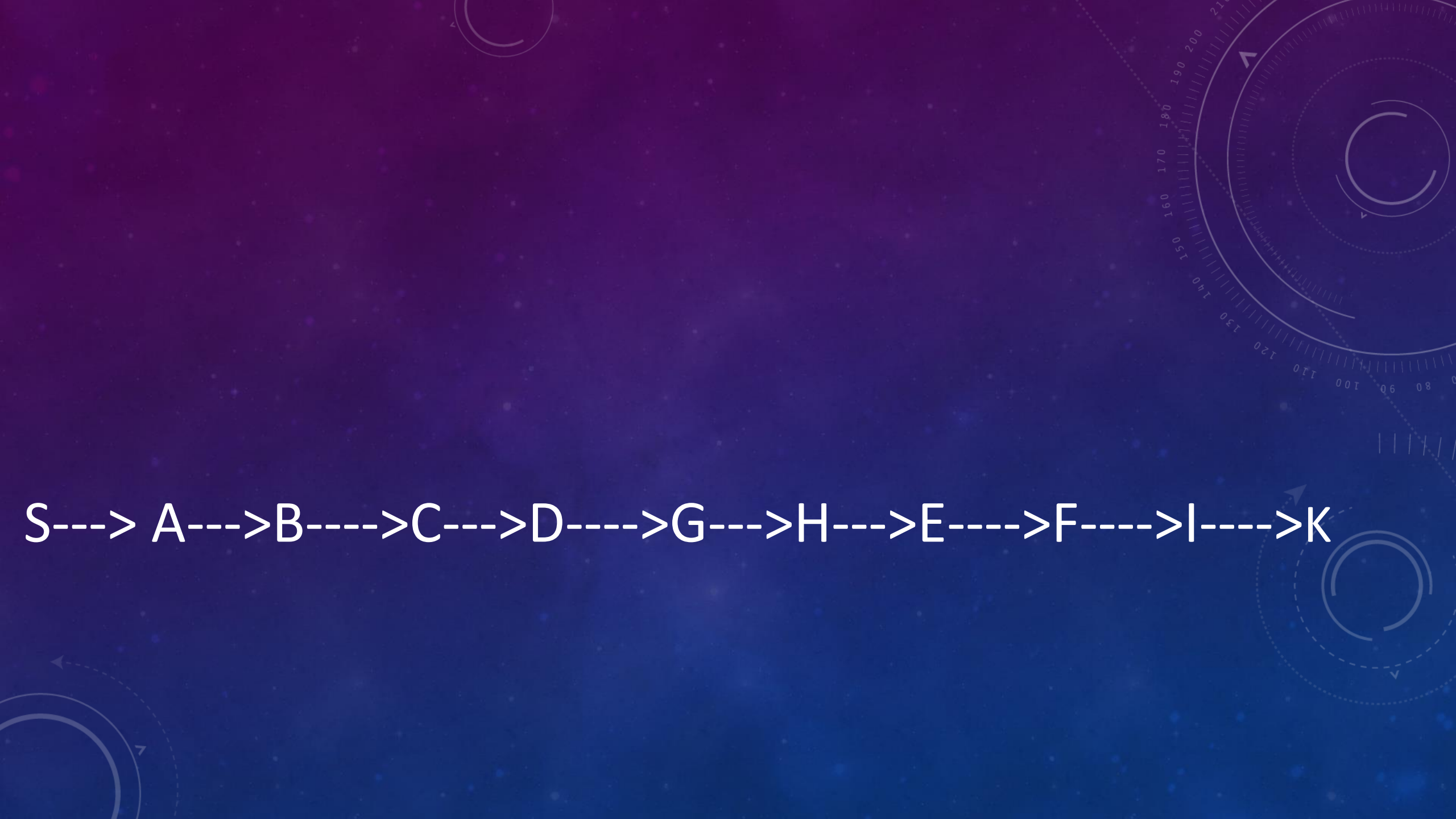
- BFS will provide a solution if any solution exists.
- If there are more than one solutions for a given problem, then BFS will provide the minimal solution which requires the least number of steps.

Disadvantages

- It requires lots of memory since each level of the tree must be saved into memory to expand the next level.
- BFS needs lots of time if the solution is far away from the root node.

Breadth First Search





S---> A--->B--->C--->D--->G--->H--->E--->F--->I--->K

Step 1: SET STATUS = 1 (ready state) for each node in G

Step 2: Enqueue the starting node A and set its STATUS = 2 (waiting state)

Step 3: Repeat Steps 4 and 5 until QUEUE is empty

Step 4: Dequeue a node N. Process it and set its STATUS = 3 (processed state).

Step 5: Enqueue all the neighbours of N that are in the ready state (whose STATUS = 1) and set their STATUS = 2(waiting state)

[END OF LOOP]

Step 6: EXIT

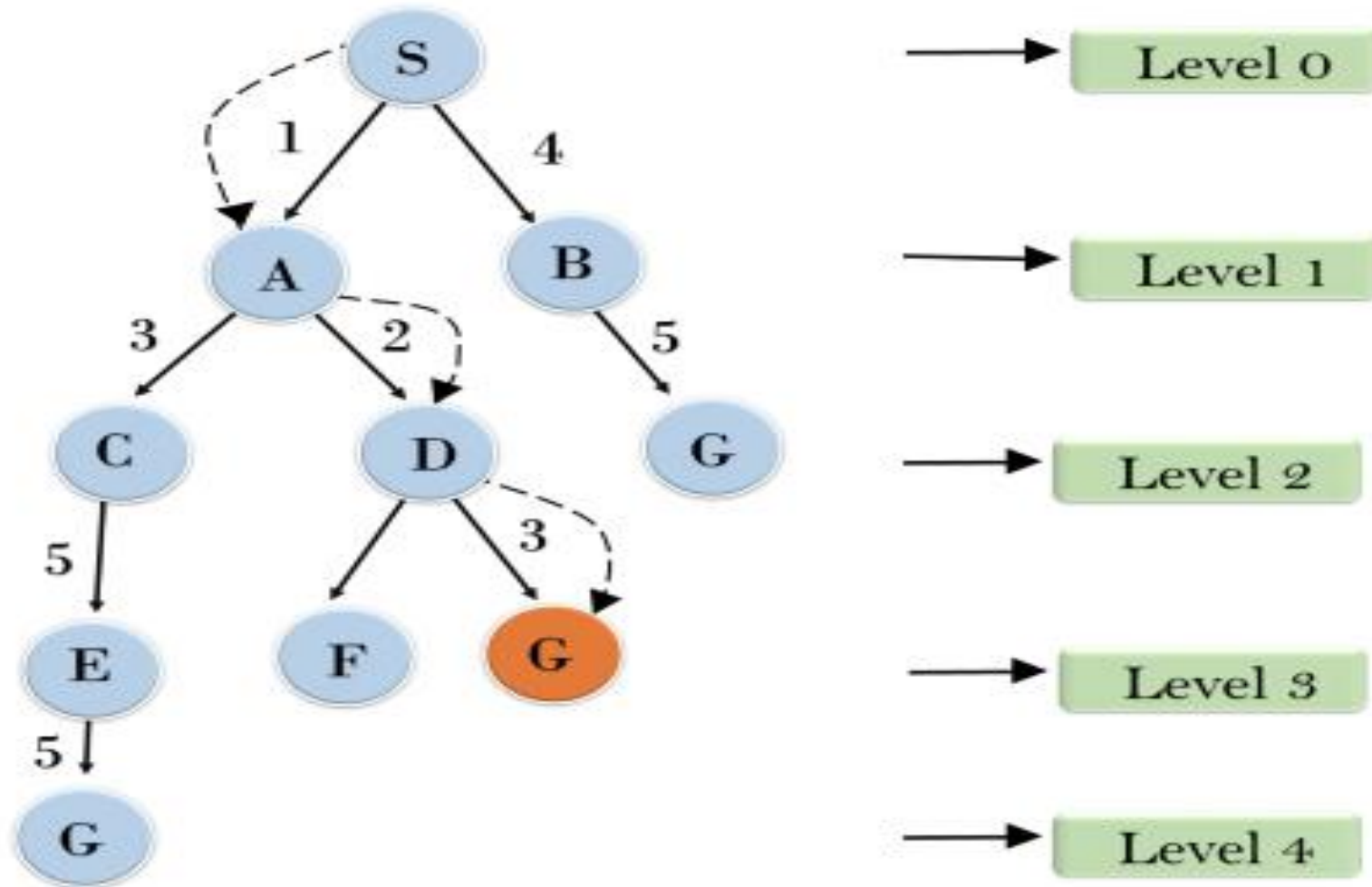
DEPTH-FIRST SEARCH (DFS)

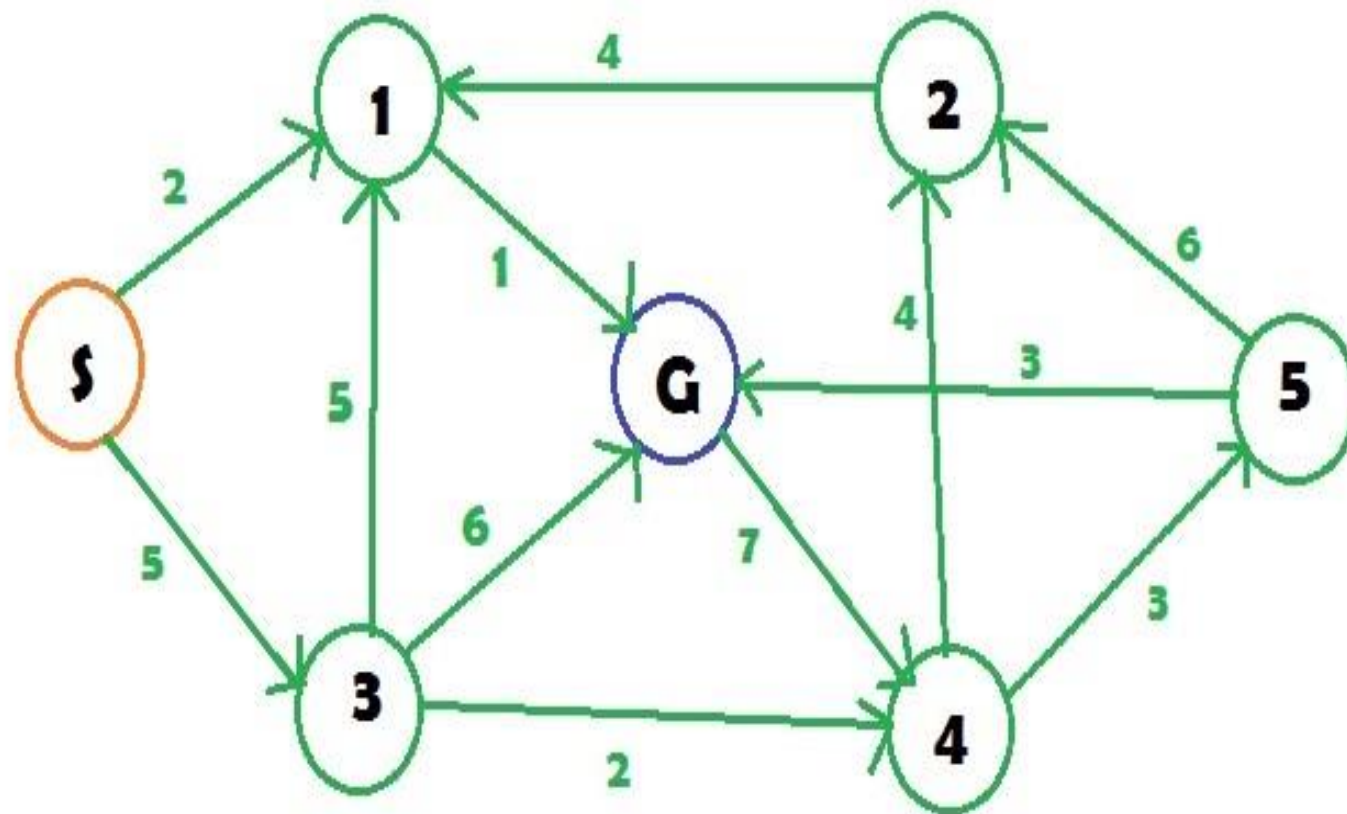
Depth-first search (DFS) is a recursive algorithm that searches through tree or graph data structures. It's a backtracking algorithm that explores as far as possible along each branch before backtracking

UNIFORM-COST SEARCH ALGORITHM

Uniform-cost search is a searching algorithm used for traversing a weighted tree or graph. This algorithm comes into play when a different cost is available for each edge. The primary goal of the uniform-cost search is to find a path to the goal node which has the lowest cumulative cost. Uniform-cost search expands nodes according to their path costs from the root node. It can be used to solve any graph/tree where the optimal cost is in demand. A uniform-cost search algorithm is implemented by the priority queue. It gives maximum priority to the lowest cumulative cost. Uniform cost search is equivalent to BFS algorithm if the path cost of all edges is the same.

Uniform Cost Search

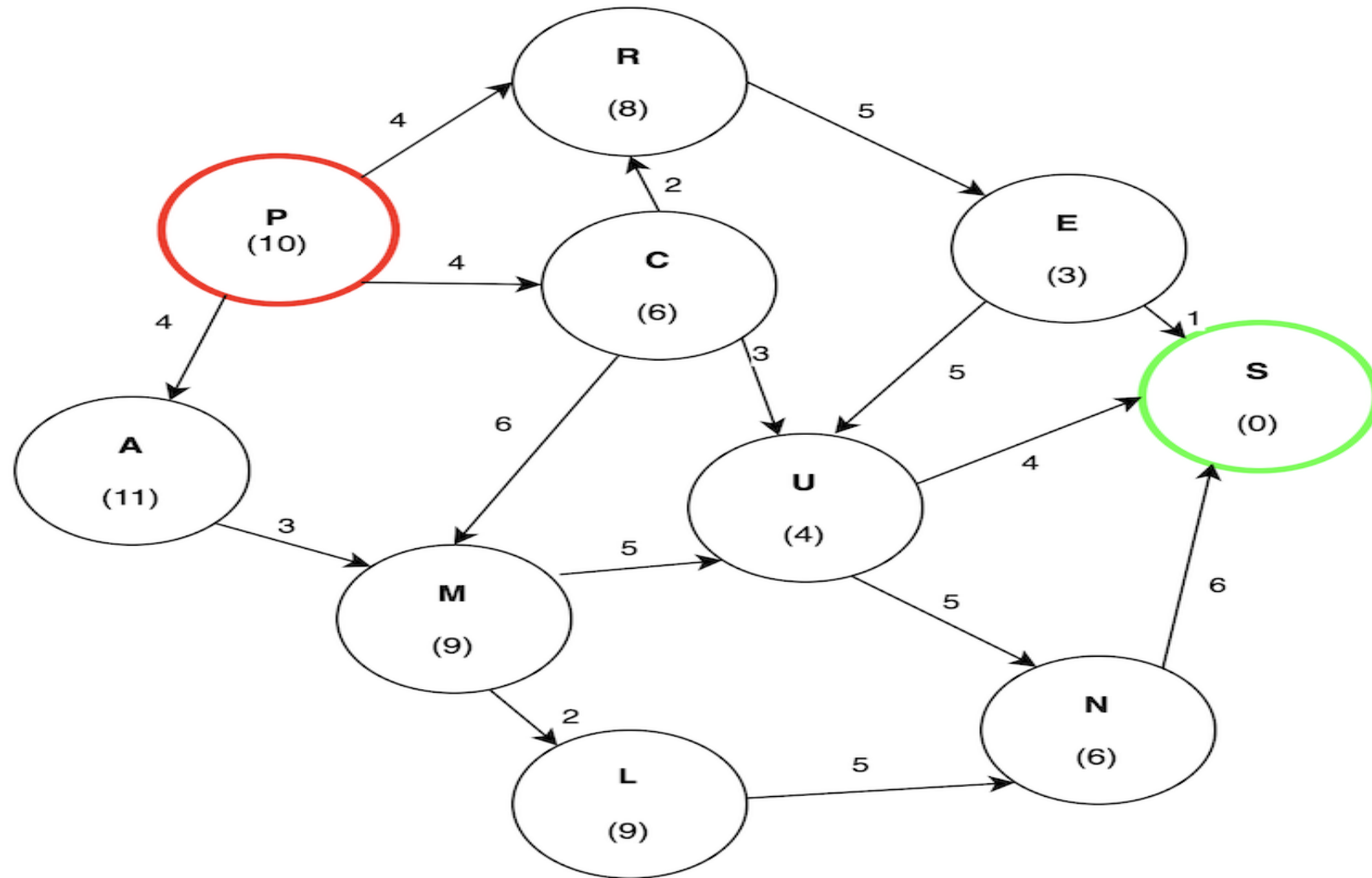




S is the starting state
G is the goal state

BEST-FIRST SEARCH ALGORITHM (GREEDY SEARCH)

- Greedy best-first search algorithm always selects the path which appears best at that moment. It is the combination of depth-first search and breadth-first search algorithms. It uses the heuristic function and search. Best-first search allows us to take the advantages of both algorithms. With the help of best-first search, at each step, we can choose the most promising node. In the best first search algorithm, we expand the node which is closest to the goal node and the closest cost is estimated by heuristic function



A* SEARCH ALGORITHM

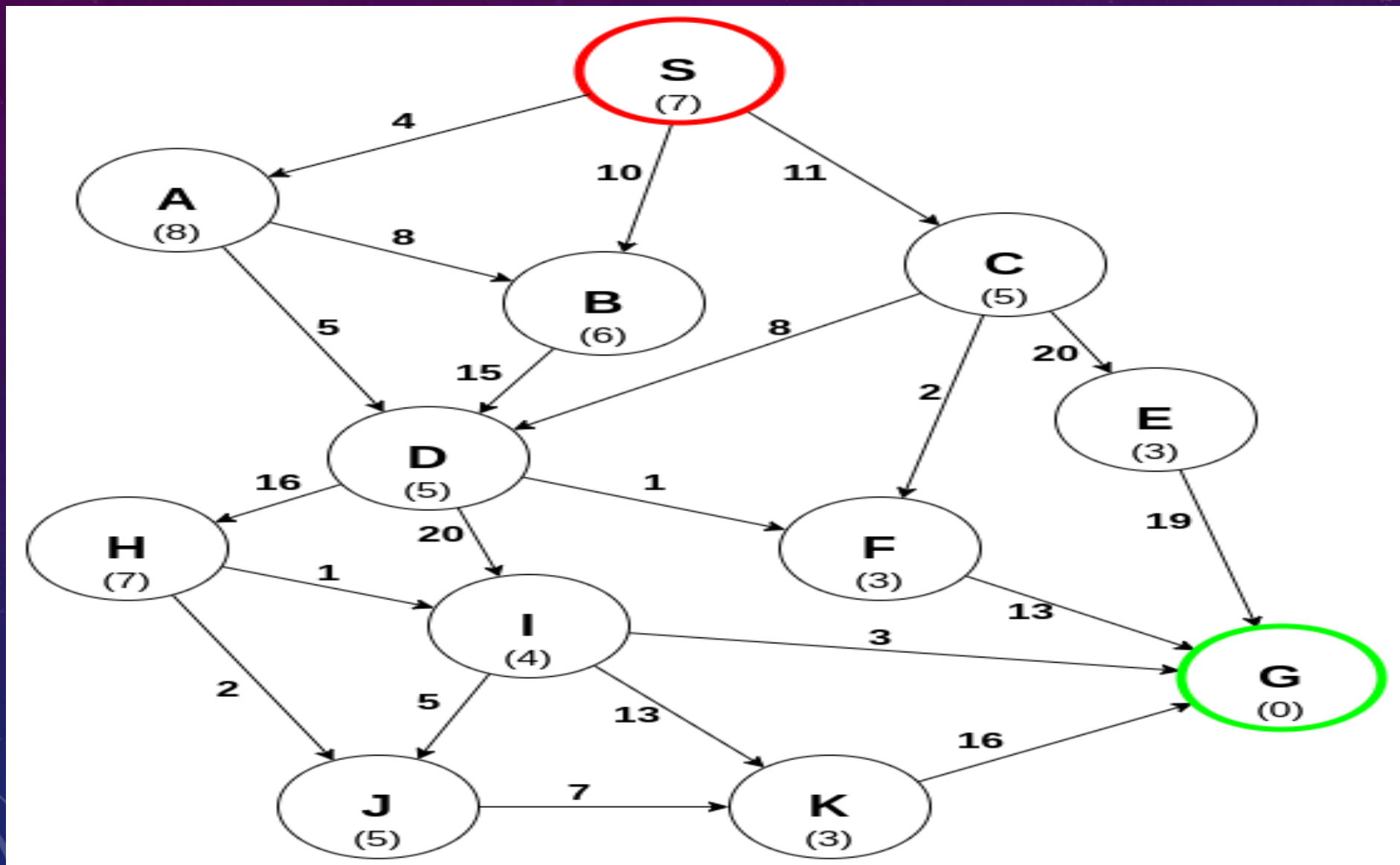
- A* search is the most commonly known form of best-first search. It uses heuristic function $h(n)$, and cost to reach the node n from the start state $g(n)$. It has combined features of UCS and greedy best-first search, by which it solve the problem efficiently. A* search algorithm finds the shortest path through the search space using the heuristic function. This search algorithm expands less search tree and provides optimal result faster. A* algorithm is similar to UCS except that it uses $g(n)+h(n)$ instead of $g(n)$.

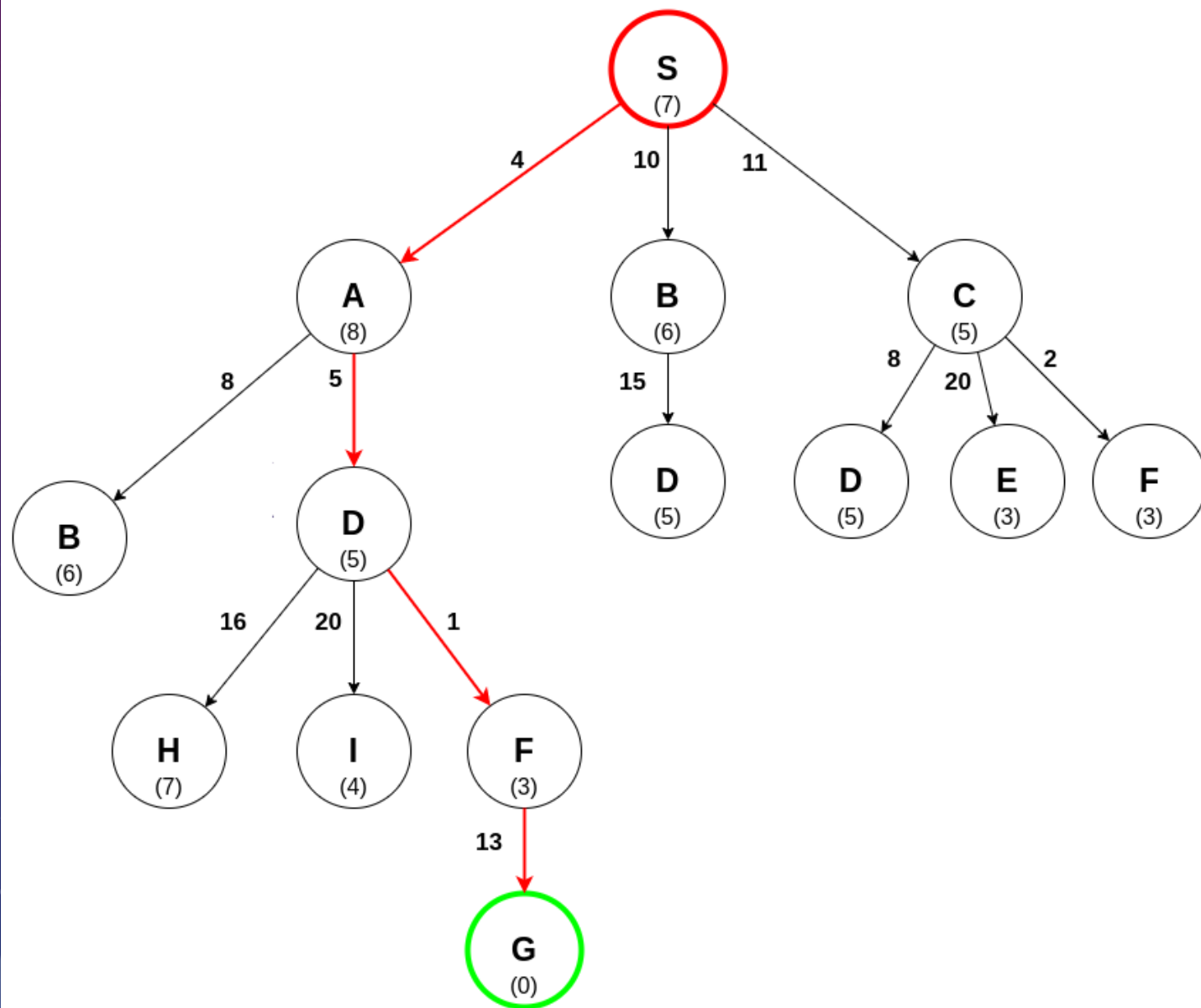
$$f(n) = g(n) + h(n)$$

Estimated cost
of the cheapest
solution.

Cost to reach
node n from
start state.

Cost to reach
from node n to
goal node





Node[cost]
I[33]
H[32]
E[36]

Closed List
S
A
D
F
B
C
G

ASSIGNMENT 1

1. Explain problem solving in AI
2. Explain different AI techniques.
3. What is an intelligent agent. Explain different categories of Intelligent agents.
4. Write and explain A* algorithm with example.
5. What are the different applications of Artificial intelligence.